

Lesson Plan — Session 1 · Customary Conversions (feet → inches)

Mr. Neft · Grade 6 Mathematics · Monday, October 5, 2026 · 90-minute block

Reveal Math: Ratio Reasoning: Convert Measurements within the Same System · Source: 3.6.pptx slides 1–21 and 39 · Practice/3.6 Session 1.docx · HW/3.6 Session 1 HW.docx

1. DO NOW

Do Now — Warm-Up (Slide 1) · 4 minutes

Students work silently, then mark one answer they are unsure of.

- 1. $12 \times 4 =$ _____
- 2. 1 foot = _____ inches
- 3. $1,000 \times 3 =$ _____
- 4. $60 \div 12 =$ _____

Talk stem (TWR)

“Item ____ will help me today because 1 foot = ____ inches.”

Key: $48 \cdot 12 \cdot 3,000 \cdot 5$

2. STANDARDS

Grade 5 Prerequisite	Grade 6 Target Standard	Grade 7 Extension
5.MD.A.1 — Convert among different-sized standard measurement units within a given measurement system (e.g., 5 cm to 0.05 m) and use these conversions in multi-step, real-world problems.	Maryland 6.AT.3 (deck, slides 3 and 24) · CCSS 6.RP.A.3d (HW and Answer Key) — Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities. Deck target (slide 39): <i>I can use ratio reasoning to convert between units of the same measurement system.</i>	7.RP.A.1 — Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units.

3. OBJECTIVES

Objective	Statement	Self-check
Content	I can convert feet to inches by scaling the unit ratio 12 inches : 1 foot in a ratio table, and use the result to compare two heights.	☐ Before ☐ After
Language (WIDA)	I will explain why the number changes but the height does not, using the frame “The units changed from ____ to ____, but the amount stayed the same” to justify my conversion to a partner. Sentence Frame: “I know this because...”	☐ Before ☐ After

4. LESSON PROCEDURES

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Warm-Up (8 min)	• Display the Warm-Up (Slide 1). Four items solo; students star one item they are unsure of. [T3] • Circulate and note who writes 1 foot = 10 inches or who multiplies on item 4 ($60 \div 12$). • Show More or Less Than 10 (Slide 12): each student picks one sum and defends an estimate. All three are less than 10 (9.53, 9.243, 9.803). [T4★] • Go over the answers $48 \cdot 12 \cdot 3,000 \cdot 5$ and ask which item helps change feet to inches.	• Solve items 1–4 alone; star one unsure answer and revise it in a second color. • Defend an estimate to a partner: <i>"I think $4.509 + 5.294$ is ___ than 10 because $4.5 + 5.3$ is about ___."</i> • Name the link: <i>"Item ___ helps because 1 foot = ___ inches."</i>	• R.M.G. — visual schedule: today's 8-phase agenda posted and pointed to before starting. • J.L. — clear expectations: "4 problems · 4 minutes" timer visible. • A.M.M. — wait time; teacher checks item 2 privately before whole-class sharing.	• L1–2: Point to 12 on a ruler to show 1 foot = 12 inches; answer items with numbers only. • L3–4: Answer and say "I multiplied/divided because ___." • L5: Explain which Warm-Up item connects to changing feet into inches.	• [T4] Scan items 2 and 4 on every paper; tally who has 1 ft = 12 in and $60 \div 12 = 5$. • [T3] "How many inches are in 1 foot? How do you know?"
Launch (7 min)	• Show the Learning Target (Slide 3); students read the content and language objectives chorally. [T3] • Run Vocabulary 1–4 (Slides 4–7): convert measurements within the same system, conversion factor, unit ratio, convert. For each: "Where does this word appear in the example?" [T4] • Have students write today's conversion factor, 1 ft = 12 in, at the top of the note sheet.	• Restate the target to a partner: "Today I will convert ___ to ___ and explain why the ___ stays the same." - Sentence Frame: "I know this because..." • Point to each word in its example: <i>"The conversion factor is 1 foot = ___ inches, so the unit ratio of inches to feet is ___ : 1."</i>	• J.C. — pre-teaching: convert / conversion factor / unit ratio cards previewed before class. • M.E. — word bank (convert, conversion factor, unit ratio, customary) on desk. • S.D.L.P. — verbal response: says each definition to the teacher instead of writing it.	• L1–2: Match word cards to pictures (ruler, 12 : 1); repeat each term aloud. • L3–4: Use the conversion-factor frame for each term. • L5: Explain how a conversion factor and a unit ratio are different. Sentence Frame: "I know this because..."	• [T4] Listen to partner restatements for "the amount (height) stays the same." • [T3] "What is the unit ratio of inches to feet?" (12 : 1)
Be Curious (10 min)	• Show Be Curious (Slide 11): 30 seconds of quiet looking at the bag on the scale, then collect "What is the question?" ideas. [T3] • Launch Riding the Attractions (Slide 13): the sign reads 44 in.; the child is 4 feet tall. Students predict with the slide frame. [T4★] • A/B: A states a prediction; B asks, "Which unit will you use?" Switch roles. • Do not confirm the answer. Record 2–3 predictions (for example, "4 is less than 44, so no") to revisit in Guided Practice.	• Write one question about the photo: <i>"I wonder how many ___ the bag weighs if the scale shows ___."</i> • Predict (Slide 13 frame): <i>"I need to compare ___ and ___ in the same unit. I can start with ___."</i> • Ask the partner for evidence: <i>"Why did you change ___ into ___?"</i>	• R.B. — simplified language card: "Child: 4 feet. Sign: 44 inches. Same unit?" • J.L. — movement: students stand at the classroom height chart to show 4 feet. • A.M.M. — safe response: predictions go on unsigned sticky notes.	• L1–2: Point to "44 in." and "4 feet"; sort them as same unit or different units. • L3–4: Use the Slide 13 prediction frame. • L5: Predict and justify why comparing 4 to 44 is not a fair comparison.	• [T4] Sort the sticky notes and count predictions that compare 4 directly to 44. • [T3] "What units are in this problem?" (feet and inches)

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Guided Practice (15 min)	- Reveal sequence (Slides 19–20): gather attempts before each reveal. Units are different; 1 foot = 12 inches. [T3] - Collaborate and Connect (Slide 21): the ratio of inches to feet is 12 : 1, a unit ratio. “How can you use the unit ratio to convert 4 feet to inches?” A explains, B asks for evidence. [T4★] - Feet → Inches (Slide 14): build the ratio table 1 : 12, 2 : 24, 4 : 48; scale both rows by 4. Name the tools with Vocabulary 6–7 (Slides 9–10). - Solution (Slide 15): 48 inches > 44 inches, so the child can ride. Return to the recorded predictions. Copy the 3 note lines (Slide 2), cover one, and explain it.	- Complete the note-sheet ratio table: Feet 1, 2, 3, 4 → Inches 12, __, __, __. - Explain the scaling (Slide 14 frame): “I multiplied 1 foot and 12 inches by __. The height stayed __.” - Prove the answer: “4 feet = __ inches, and __ inches is more than 44 inches, so the child __ ride.”	- S.D.L.P. — manipulatives: four 12-inch rulers laid end to end to model 4 feet = 48 inches. - R.B. — graphic organizer: pre-printed two-row ratio table with unit labels. - J.C. — extended time for note copying; Slide 2 lines provided as a printed strip.	- L1–2: Fill the ratio table with number cards; say “4 feet equals 48 inches.” - L3–4: Use the Slide 14 frame. - L5: Explain why multiplying both quantities by 4 keeps an equivalent ratio. Sentence Frame: “I know this because...”	[T4] Check every table for 24 and 48 in the inches row, aligned under 2 and 4. [T3] “Why do we multiply both 1 and 12 by 4, not just one of them?”
Collaborative Practice (15 min)	- Practice: Customary Conversions (Slide 16): 5 quarts to fluid ounces (1 qt = 32 fl oz), then write ounces : pounds and pounds : ounces. [T4★] - Assign Practice sheet items 2–4 (Field Day Ribbon, Party Punch, Gym Rope) A/B: A computes; B checks a second way. - Discuss the task (Slide 17): “Why does the number change while the actual measurement stays the same?” [T4] - Sentence Frame: “I know this because...”	- Slide 16: $5 \times 32 = 160$ fl oz; “The ratio __ compares ounces to pounds. Reversing the order gives __.” (16 : 1 → 1 : 16) - Items 2–4 (15 ft, 180 in · 12 qt, 48 cups · 24 ft, 288 in). Partner check: “1 yard is __ inches, so 5 yards is $5 \times \text{__} = \text{__}$ inches.” - Slide 17: “The number got bigger because inches are __ than feet, but the length stayed __.”	- J.L. — movement: partners switch seats between items 2 and 3. - R.B. — graphic organizer: arrow chain yd → ft → in labeled $\times 3$ and $\times 12$. - S.D.L.P. — calculator for the two-step products ($5 \times 3 \times 12$; $8 \times 3 \times 12$).	- L1–2: Partner reads the problem aloud; student fills the table cells and labels the units. - L3–4: Use the partner-check frame. - L5: Explain why two steps ($\times 3$, then $\times 12$) equal one step ($\times 36$). Sentence Frame: “I know this because...”	[T4] Listen for B partners checking with 36 inches per yard. [T3] “How many cups are in 1 gallon?” ($4 \times 4 = 16$)
Independent Practice (12 min)	- Launch the three stations in Section 7. Post the rotation chart before releasing students. - Lead the Reteach table (Practice items 1, 10, 11) with rulers and the ratio table. [T4] - At minute 6, check the Practice station’s item 7 (24 quarts → gallons), the first item that requires dividing.	- Work the assigned station set; label units on every step. - Before each item: “__ are smaller than __, so the number should get __, so I __.”	- M.E. — extended time with a core set (items 1, 10, 11); sentence starters on the table. - A.M.M. — 1:1 check-in after the first item; reassurance before moving on. - R.M.G. — visual station schedule and predictable rotation signal.	- L1–2: Items marked with a big-to-small arrow picture; answer with number and unit. - L3–4: Use the direction frame before each item. - L5: Item 13 — explain Sam’s error in writing.	[T4] Check the two dividing items (7 and 11) for every student at the Practice station. [T3] “Is a pound bigger or smaller than an ounce? So will 80 get bigger or smaller?”

Phase (Time)	Teacher Moves	Student Moves	SPED Supports	Level 1 / WIDA Scaffolds	Formative Check
Closing (8 min)	60-Second Check (Slide 18): a child is 5 feet tall; convert to inches, explain, and self-monitor (READY / ALMOST / NOT YET). [T4★] - Learning Targets (Slide 39): students rate themselves on the first target. - Assign HW/3.6 Session 1 HW; preview Session 2 (liters and milliliters).	- Convert 5 ft = 60 in with a ratio table or equation. - Explain: <i>"The number changed from ____ to ____ because each foot holds ____ inches, but the height stayed the same."</i> - Circle READY, ALMOST, or NOT YET and ask a partner, "How do you know?"	- J.C. — verbal response for the explain part. - J.L. — reinforcement: specific praise for a revised answer. - R.M.G. — predictability: Session 2 agenda shown before dismissal.	- L1–2: Show the answer on a whiteboard; point to READY, ALMOST, or NOT YET. - L3–4: Use the explain frame. - L5: Explain to the class, pointing to the ratio table as evidence.	[T4] Sort 60-Second Checks into READY / ALMOST / NOT YET and compare each self-rating to the work. [T3] "What do you multiply by to change feet to inches?" (12)
Exit Ticket (15 min)	- Assign exit-ticket tiers (Section 9) using the 60-Second Check piles. - Silent and independent; read aloud on request. - Collect and sort to form Session 2 Reteach and Enrichment groups. [T4]	- Complete the assigned tier; show the conversion and label units. - Tier 1 frame: <i>"I ____ by 12 because inches are ____ than feet. 6 feet = ____ inches."</i>	- M.E. — extended time and word bank. - S.D.L.P. — calculator. - J.C. — text-to-speech or teacher read-aloud. - A.M.M. — 1:1 check-in and reassurance before submitting.	- L1–2: Tier 1 read aloud with a ruler picture. - L3–4: Tier 2 with the explain frame. - L5: Tier 3.	[T4] Score each ticket against the Section 9 key. [T3] On collection: "Did you multiply or divide? Why?" Target: 80%+ convert 6 ft = 72 in and state they multiplied because inches are smaller than feet.

Total: 90 minutes. TEACH codes: **[T3]** ask–respond–affirm · **[T4]** justify · **[T4★]** justify + peer discourse or written reasoning.

5. VOCABULARY

Tier	Term	Student-Friendly Definition	Morphology (root + affix)	EN-ES Cognate ✓/✗	Cross-Disciplinary Use
3	customary units	The U.S. units: inch, foot, yard, mile, ounce, pound, cup, quart, gallon.	custom + -ary (relating to)	✗ (unidades usuales)	Cooking and recipes (cups, ounces); U.S. road signs (miles)
2	convert	Write the same amount using a different unit.	con- (together) + vert (turn)	✓ convertir	Science: convert grams to kilograms; money: convert dollars to cents
3	conversion factor	The pair of equal amounts you multiply or divide by, such as 1 ft = 12 in.	convert + -sion; fact (make) + -or	✓ factor de conversión	Science lab measurement; currency exchange rates
3	unit ratio	A ratio that compares an amount to exactly 1 unit, like 12 inches to 1 foot.	unit (Latin unus, one) + ratio (reckoning)	✓ razón unitaria (unitaria ✓)	Unit price in shopping; speed in miles per hour
3	ratio table	A table of equal ratios that scales a conversion up or down one column at a time.	ratio (reckoning) + table	✓ tabla de razones	Recipes scaled for more people; science data tables
2	measurement system	A family of units that go together — customary or metric.	measure + -ment; syn- (together) + stem (stand)	✓ sistema de medición	Science (metric); geography (map scales)

6. MISCONCEPTIONS + INTERVENTIONS

#	Student Belief	Cognitive Root	Teacher Corrective Move	Intervention Route
1	Converting inches to feet, multiply by 12 (Sam: $60 \times 12 = 720$ feet).	Memorized “use 12” without reasoning about unit size; no reasonableness check.	Ask: “Are feet bigger or smaller than inches? Should there be more or fewer of them?” Model 60 in on rulers $\rightarrow$ 5 groups of 12.	Practice item 13 (Sam’s Work); Reteach station item 11 (80 oz $\rightarrow$ lb).
2	40 inches is more than 4 feet because $40 > 4$.	Compares the numbers and ignores the units.	Return to Riding the Attractions (Slides 19, 13): change to the same unit first, then compare 48 and 40.	Slide 19 revisit; HW item 8 (Elijah’s ribbon).
3	Scale only one quantity in the ratio table (1 : 12 $\rightarrow$ 4 : 12) or add instead of multiply.	Treats the table as two separate lists, not paired equivalent ratios.	Point to Slide 14: circle a column and say “both rows times 4.” Check with rulers: 4 rulers = 48 in.	Slide 14 table; Reteach station item 1b (Roller Coaster table).
4	ounces : pounds and pounds : ounces are the same ratio.	Order of quantities in a ratio is not yet attached to meaning.	Label each number with its unit; read aloud: “16 ounces for every 1 pound” vs “1 pound for every 16 ounces.”	Slide 16 item 2; Collaborative Practice.
5	Stops after one step in a two-step conversion (5 yd = 15, done).	Loses track of the target unit across two conversion factors.	Write the unit chain yd $\rightarrow$ ft $\rightarrow$ in above the work and check off each arrow.	Practice items 2, 4, 12; Practice-station partner check.

7. SMALL-GROUP STATIONS (RUN DURING INDEPENDENT PRACTICE)

Station	Who	Problems	Guiding Question / Protocol	Expected Artifact	TWR Frame
RETEACH (teacher-led)	M.E., R.B., J.C., R.M.G., J.L., S.D.L.P., A.M.M. (full accommodations) + students who placed NOT YET on the Warm-Up	Practice sheet item 1 (Roller Coaster Height Limit: 48 in rule, Caleb 4 ft) · item 10 (Bookshelf 6 ft $\rightarrow$ 72 in) · item 11 (Bag of Flour 80 oz $\rightarrow$ 5 lb)	“Is the new unit bigger or smaller? So will the number grow or shrink?”	Labeled ratio table for item 1 (4 ft = 48 in, Caleb can ride) plus two direction-labeled conversions.	“Feet are ___ than inches, so I ___ by 12. 4 feet = ___ inches.”
PRACTICE (partner)	Partners (A/B), mixed READY / ALMOST	Practice sheet items 5–9 and 12 (4.5 lb $\rightarrow$ 72 oz · 2.5 mi $\rightarrow$ 13,200 ft · 24 qt $\rightarrow$ 6 gal · 6.5 lb $\rightarrow$ 104 oz · 9 gal $\rightarrow$ 36 qt $\rightarrow$ 72 pint cups · 7 yd $\rightarrow$ 21 ft $\rightarrow$ 252 in)	A solves; B checks the direction (multiply or divide) before checking the number. Switch every item.	Each item answered with units, plus B’s initials beside every check.	“You should ___ because ___ are bigger than ___, so there will be ___ of them.”
ENRICHMENT (independent)	Independent, READY on the Warm-Up and 60-Second Check	Practice item 13 (Sam’s Work) · item 14 (Fencing a Garden: 24 ft $\rightarrow$ 8 yd, \$72; by the foot $24 \times \$3.25 = \78)	Why is buying by the foot more expensive, even though the fence is the same length?	A written paragraph comparing the two prices with labeled conversions.	“Buying by the yard costs ___ because 24 feet = ___ yards. At \$3.25 per foot, 1 yard costs ___, which is ___ than \$9.”

8. STUDENT NOTE SHEET

Lesson 3-6 • Converting Within One System • Session 1 Notes Name _____ Date _____ Period _____

A • Warm-Up Link

1 foot = _____ inches

$60 \div 12 = \underline{\hspace{2cm}}$ This helps me change _____ into _____.

B • Riding the Attractions (Slides 13–15)

Sign: YOU MUST BE THIS TALL TO RIDE — 44 in. Child: 4 feet

My prediction: “I need to compare _____ and _____ in the same unit. I can start with _____.”

Feet	1	2	3	4
Inches	12	_____	_____	_____

4 feet = _____ inches. _____ inches > 44 inches, so the child _____ ride.

C • My 3 Lines (Slide 2)

01 Convert means write the _____ amount in another _____.

02 Start with a known _____.

03 Use _____ to find the new number.

D • Which Way? (Slide 16)

Bigger unit → smaller unit: _____ (5 quarts = _____ fluid ounces, 1 qt = 32 fl oz)

Smaller unit → bigger unit: _____ (80 ounces = _____ pounds, 1 lb = 16 oz)

E • Explain It (TWR)

“A foot is _____ than an inch, so for the same height the number of inches is always _____ than the number of feet, because _____.”

Goal tracker: I can convert customary units using a unit ratio and explain why the amount stays the same.

Sentence Frame: “I know this because...”

☐ 1 — Need help

☐ 2 — Need a hint

☐ 3 — Got it

☐ 4 — Can teach it

9. EXIT TICKET

Tier	Task	Frame	Answer Key
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Tier	Task	Frame	Answer Key
Tier 1 · Identify + frame	Convert 6 feet to inches. (1 foot = 12 inches) Circle: multiply / divide.	"I ___ by 12 because inches are ___ than feet. 6 feet = ___ inches."	Multiply · smaller · 72 inches.
Tier 2 · Identify + explain (TWR 2–3 sentences)	Adapted from Riding the Attractions (Slide 13): A ride requires riders to be at least 4 feet tall. Ana is 50 inches tall. Can she ride? Show the conversion, then explain in 2–3 sentences.	"First I changed ___ to ___ because _____. Then I compared ___ and _____. So Ana ___ ride."	$4 \times 12 = 48$ inches; $50 > 48$, so yes, Ana can ride. The explanation must change to one unit before comparing.
Tier 3 · Multi-step + contrastive + transfer	Banner Fabric (Practice item 23): A banner needs 7 feet of fabric. Fabric is sold by the yard (3 ft = 1 yd). (a) Write 7 feet in yards. (b) How many whole yards must you buy? (c) Jordan says buy 2 yards because 2 $\frac{1}{3}$ rounds to 2. Explain who is right, using feet as evidence. (d) Transfer: How would your method change for 7 pints of juice sold only by the quart (2 pt = 1 qt)?	"Jordan is ___ because 2 yards is only ___ feet, which is ___ than 7 feet."	(a) $7 \div 3 = 2 \frac{1}{3}$ yd. (b) 3 yards. (c) Jordan is wrong: $2 \text{ yd} = 6 \text{ ft} < 7 \text{ ft}$. (d) Same move, dividing by 2: $7 \div 2 = 3 \frac{1}{2}$ qt, so buy 4 quarts.

10. GRADE 6 IEP ACCOMMODATIONS MATRIX

Student	Full Accommodation List	Applied This Session (phase: support)	Small Group
M.E.	Sentence starters, word banks, chunking, extended time, manipulatives	- Launch: word bank (convert, conversion factor, unit ratio, customary) on desk. - Independent Practice: extended time with a core set (items 1, 10, 11); sentence starters on the table. - Exit Ticket: extended time and word bank.	Reteach station (all 7 students)
R.B.	Graphic organizers, simplified language, small group, breaks	- Be Curious: simplified language card: "Child: 4 feet. Sign: 44 inches. Same unit?" - Guided Practice: graphic organizer: pre-printed two-row ratio table with unit labels. - Collaborative Practice: graphic organizer: arrow chain yd $\rightarrow$ ft $\rightarrow$ in labeled $\times 3$ and $\times 12$.	Reteach station (all 7 students)
J.C.	Text-to-speech, verbal responses, extended time, pre-teaching	- Launch: pre-teaching: convert / conversion factor / unit ratio cards previewed before class. - Guided Practice: extended time for note copying; Slide 2 lines provided as a printed strip. - Closing: verbal response for the explain part. - Exit Ticket: text-to-speech or teacher read-aloud.	Reteach station (all 7 students)
R.M.G.	Visual schedules, sensory mods, motor mods, predictability	- Warm-Up: visual schedule: today's 8-phase agenda posted and pointed to before starting. - Independent Practice: visual station schedule and predictable rotation signal. - Closing: predictability: Session 2 agenda shown before dismissal.	Reteach station (all 7 students)
J.L.	Movement, clear expectations, reinforcement, chunking, visuals	- Warm-Up: clear expectations: "4 problems · 4 minutes" timer visible. - Be Curious: movement: students stand at the classroom height chart to show 4 feet. - Collaborative Practice: movement: partners switch seats between items 2 and 3. - Closing: reinforcement: specific praise for a revised answer.	Reteach station (all 7 students)
S.D.L.P.	Multi-sensory, manipulatives, verbal responses, calculator	- Launch: verbal response: says each definition to the teacher instead of writing it. - Guided Practice: manipulatives: four 12-inch rulers laid end to end to model 4 feet = 48 inches. - Collaborative Practice: calculator for the two-step products ($5 \times 3 \times 12$; $8 \times 3 \times 12$). - Exit Ticket: calculator.	Reteach station (all 7 students)
A.M.M.	Wait time, safe responses, 1:1 check-ins, reassurance	- Warm-Up: wait time; teacher checks item 2 privately before whole-class sharing. - Be Curious: safe response: predictions go on unsigned sticky notes. - Independent Practice: 1:1 check-in after the first item; reassurance before moving on. - Exit Ticket: 1:1 check-in and reassurance before submitting.	Reteach station (all 7 students)